

EXTRA LAB PRACTISE FOR DATABASE CONCEPTS

Introduction to SQL LAB EXERCISES:

Lab 1 : Create a database called library_db and a table books with columns: book_id, title, author, publisher, year_of_publication, and price. Insert five records into the table.

```
mysql> create database library_db;  
Query OK, 1 row affected (1.595 sec)
```

```
mysql> use library_db;  
Database changed
```

```
mysql> create table books(  
-> book_id int primary key ,  
-> title varchar(20) not null,  
-> author varchar(20),  
-> publisher varchar(20),  
-> year_of_publisher varchar(10),  
-> price int not null);  
Query OK, 0 rows affected (1.588 sec)
```

```
mysql> desc books;
```

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	
titles	varchar(30)	YES		NULL	
author	varchar(20)	YES		NULL	
publish	varchar(30)	YES		NULL	
year_of_publisher	varchar(10)	YES		NULL	
price	int	NO		NULL	

```
6 rows in set (0.516 sec)
```

Lab 2: Create a table members in library_db with columns: member_id, member_name, date_of_membership, and email. Insert five records into this table.

```
mysql> create table member(  
-> member_id int primary key,  
-> member_name varchar(30),  
-> date_of_membership date,  
-> email varchar(30));  
Query OK, 0 rows affected (0.431 sec)
```

```
mysql> desc member;
```

Field	Type	Null	Key	Default	Extra
member_id	int	NO	PRI	NULL	
member_name	varchar(30)	YES		NULL	
date_of_membership	date	YES		NULL	
email	varchar(30)	YES		NULL	

```
4 rows in set (0.047 sec)
```

SQL Syntax LAB EXERCISES:

Lab 1 : Retrieve all members who joined the library before 2022. Use appropriate SQL syntax with WHERE and ORDER BY.

```
mysql> select * from member where date_of_membership < "2022-12-30" order by member_name;
```

member_id	member_name	date_of_membership	email
206	kavita desai	2021-12-30	kavitadesai@gamil.com
204	neha singh	2020-11-05	nehasingh@gmail.com
202	priya patel	2021-06-10	priyapatel@gmail.com
201	rahul sharma	2022-01-15	rahulsharma@gmail.com
205	rohan mehta	2022-09-18	rohanmehta@gmail.com

```
5 rows in set (0.023 sec)
```

Lab 2: Write SQL queries to display the titles of books published by a specific author. Sort the results by year_of_publication in descending order.

```
mysql> select author from books order by year_of_publisher;
```

author
reema thareja
raghu ramkrishan
abraham silberschatz
andrew s . tanenbaum
tom m mitchell
stuart russell
gudio van rossum

```
7 rows in set (0.013 sec)
```

SQL Constraints LAB EXERCISES:

Lab 3: Add a CHECK constraint to ensure that the price of books in the books table is greater than 0.

```
mysql> ALTER TABLE books ADD CONSTRAINT price CHECK (price > 0);
Query OK, 7 rows affected (1.723 sec)
Records: 7 Duplicates: 0 Warnings: 0
```

Lab 4: Modify the members table to add a UNIQUE constraint on the email column, ensuring that each member has a unique email address.

```
mysql> alter table member change email email1 varchar(30) unique;
Query OK, 0 rows affected, 1 warning (1.120 sec)
Records: 0 Duplicates: 0 Warnings: 1
```

Main SQL Commands and Sub-commands (DDL) LAB EXERCISES:

Lab 3: Create a table authors with the following columns: author_id, first_name, last_name, and country. Set author_id as the primary key.

```
mysql> create table author(
-> author_id int primary key,
-> first_name varchar(30),
-> last_name varchar(30),
-> country varchar(30));
Query OK, 0 rows affected (0.531 sec)
```

Lab 4: Create a table publishers with columns: publisher_id, publisher_name, contact_number, and address. Set publisher_id as the primary key and contact_number as unique.

```
mysql> create table publisher(
-> publisher_id int primary key,
-> publisher_name varchar(30),
-> contact_number varchar(10) unique ,
-> address varchar(30));
Query OK, 0 rows affected (1.000 sec)
```

ALTER Command LAB EXERCISES:

Lab 3: Add a new column genre to the books table. Update the genre for all existing records.

```
mysql> alter table books add column genre varchar(30);
Query OK, 0 rows affected (0.884 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> update books set genre="education" where book_id=101;
Query OK, 1 row affected (0.187 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> update books set genre="education" where book_id=102;
Query OK, 1 row affected (0.136 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> update books set genre="education" where book_id=103;
Query OK, 1 row affected (0.060 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> update books set genre="programing" where book_id=104;
Query OK, 1 row affected (0.141 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> update books set genre="technology" where book_id=105;
Query OK, 1 row affected (0.066 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> update books set genre="programing" where book_id=106;
Query OK, 1 row affected (0.152 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> update books set genre="technology" where book_id=107;
Query OK, 1 row affected (0.148 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

Lab 4: Modify the members table to increase the length of the email column to 100 characters.

```
mysql> alter table member change email email varchar(100);
Query OK, 5 rows affected (1.278 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

DROP Command LAB EXERCISES:

Lab 3: Drop the publishers table from the database after verifying its structure

```
mysql> drop table publisher;
Query OK, 0 rows affected (0.295 sec)
```

Lab 4: Create a backup of the members table and then drop the original members table.

```
mysql> create table member2 as select * from member1;
Query OK, 5 rows affected (0.318 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> drop table member1;
Query OK, 0 rows affected (0.500 sec)
```

Data Manipulation Language (DML) LAB EXERCISES:

Lab 4: Insert three new authors into the authors table, then update the last name of one of the authors.

```
mysql> INSERT INTO books
-> (book_id, titles, author, publish, year_of_publisher, price, genre)
-> VALUES
-> (204, 'computer networks', 'andrew s tanenbaum', 'pearson', 2021, 720, 'education'),
-> (205, 'data structures and algorithms', 'thomas h cormen', 'mit press', 2020, 800, 'programming'),
-> (206, 'artificial intelligence basics', 'stuart russell', 'pearson', 2022, 900, 'technology');
Query OK, 3 rows affected (0.081 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

Lab 5: Delete a book from the books table where the price is higher than \$100.

```
mysql> delete from books where price >100;
Query OK, 16 rows affected (0.187 sec)

mysql> select * from books;
Empty set (0.013 sec)
```

UPDATE Command LAB EXERCISES:

Lab 3: Update the year_of_publication of a book with a specific book_id.

```
mysql> update books set year_of_publisher=2025 where book_id=204;
Query OK, 1 row affected (0.146 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

Lab 4: Increase the price of all books published before 2015 by 10%.

```
mysql> update books set year_of_publisher = year_of_publisher*0.1 where year_of_publisher<"2015-12-31";
Query OK, 0 rows affected (0.018 sec)
Rows matched: 0 Changed: 0 Warnings: 0
```

DELETE Command LAB EXERCISES:

Lab 3: Remove all members who joined before 2020 from the members table.

```
mysql> delete from member where date_of_membership < "2020-12-31";
Query OK, 0 rows affected (0.017 sec)

mysql> select * from member;
+-----+-----+-----+-----+
| member_id | member_name | date_of_membership | email |
+-----+-----+-----+-----+
| 1 | Amit Patel | 2023-01-15 | amit.patel@gmail.com |
| 2 | Riya Shah | 2023-03-10 | riya.shah@gmail.com |
| 3 | Karan Mehta | 2022-12-20 | karan.mehta@gmail.com |
| 4 | Neha Joshi | 2025-04-12 | neha.joshi@gmail.com |
| 5 | Rahul Verma | 2023-09-18 | rahul.verma@gmail.com |
+-----+-----+-----+-----+
5 rows in set (0.015 sec)
```

Lab 4: Delete all books that have a NULL value in the author column.

```
mysql> delete from books where author="NULL";
Query OK, 0 rows affected (0.012 sec)
```

Data Query Language (DQL) LAB EXERCISES:

Lab 4: Write a query to retrieve all books with price between \$50 and \$100.

```
mysql> select * from books where price between 50 and 100;
Empty set (0.007 sec)
```

Lab 5: Retrieve the list of books sorted by author in ascending order and limit the results to the top 3.

```
mysql> select * from author order by first_name,last_name limit 3;
+-----+-----+-----+-----+
| author_id | first_name | last_name | country |
+-----+-----+-----+-----+
|          2 | abraham   | silberschatz | usa     |
|          3 | andrew    | tanenbaum   | netherlands |
|          4 | guido     | van rossum   | netherlands |
+-----+-----+-----+-----+
3 rows in set (0.110 sec)
```

Data Control Language (DCL) LAB EXERCISES:

Lab 3: Grant SELECT permission to a user named librarian on the books table.

```
mysql> grant select
-> on books
-> to root;
ERROR 1142 (42000): GRANT command denied to user 'root'@'localhost' for table 'books'
mysql> insert into books(book_id,titles,author,publish,year_of_publisher,price,genre) values
-> (1,'a','b','c','d',1,'f');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '(1,'a','b','c','d',1,'f');' at line 1
mysql> insert into books(book_id,titles,author,publish,year_of_publisher,price,genre) values
-> (1,'a','b','c','d',1,'f');
ERROR 1142 (42000): INSERT command denied to user 'root'@'localhost' for table 'books'
```

Lab 4: Grant INSERT and UPDATE permissions to the user admin on the members table.

```
mysql> use library_db;
Database changed
mysql> grant insert , update
-> on member
-> to admin;
ERROR 1142 (42000): GRANT command denied to user 'admin'@'localhost' for table 'member'
mysql> select * from member;
ERROR 1142 (42000): SELECT command denied to user 'admin'@'localhost' for table 'member'
```

REVOKE Command LAB EXERCISES:

- Lab 3: Revoke the INSERT privilege from the user librarian on the books table

```
ERROR 1147 (42000): There is no such grant definition for the operation specified
mysql> revoke select
-> on books
-> from librarian;
Query OK, 0 rows affected (0.049 sec)
```

- Lab 4: Revoke all permissions from user admin on the members table.

```
mysql> revoke insert , update
-> on member
-> from admin;
Query OK, 0 rows affected (0.054 sec)
```

Transaction Control Language (TCL) LAB EXERCISES:

- Lab 3: Use COMMIT after inserting multiple records into the books table, then make another insertion and perform a ROLLBACK.

```
mysql> INSERT INTO books
-> (book_id, titles, author, publish, year_of_publisher, price, genre)
-> VALUES
-> (301, 'software engineering', 'ian somerville', 'pearson', 2021, 850, 'education'),
-> (302, 'web technologies', 'jeffrey ullman', 'pearson', 2020, 700, 'technology'),
-> (303, 'introduction to algorithms', 'thomas h cormen', 'mit press', 2019, 950, 'programming');
Query OK, 3 rows affected (0.021 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> commit;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near
ommit' at line 1
mysql> commit;
Query OK, 0 rows affected (0.000 sec)

mysql> select * from books;
```

book_id	titles	author	publish	year_of_publisher	price	genre
101	database management system	raghu raskrishan	mcgraw hill	2018	550	education
102	operating system concept	abraham silberschatz	wiley	2019	650	education
103	computer network	andrew s . tanenbaum	pearson	2020	720	education
104	data structure using c	reema thareja	oxford university press	2017	480	programming
105	artificial intelligence	stuart russell	pearson	2021	900	technology
106	python programming	gudiu van rossum	o reilly media	2022	550	programming
107	machine learning basics	tom m mitchell	mcgraw hill	2020	700	technology
108	computer networks	andrew s tanenbaum	pearson	2020	720	education
205	data structures and algorithms	thomas h cormen	mit press	2020	800	programming
206	artificial intelligence basics	stuart russell	pearson	2022	900	technology
301	software engineering	ian somerville	pearson	2021	850	education
302	web technologies	jeffrey ullman	pearson	2020	700	technology
303	introduction to algorithms	thomas h cormen	mit press	2019	950	programming

13 rows in set (0.016 sec)

```
mysql> INSERT INTO books
-> (book_id, titles, author, publish, year_of_publisher, price, genre)
-> VALUES
-> (304, 'software engineering', 'ian somerville', 'pearson', 2021, 850, 'education'),
-> (305, 'web technologies', 'jeffrey ullman', 'pearson', 2020, 700, 'technology'),
-> (306, 'introduction to algorithms', 'thomas h cormen', 'mit press', 2019, 950, 'programming');
Query OK, 3 rows affected (0.135 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> rollback;
Query OK, 0 rows affected (0.000 sec)

mysql> select * from books;
```

book_id	titles	author	publish	year_of_publisher	price	genre
101	database management system	raghu raskrishan	mcgraw hill	2018	550	education
102	operating system concept	abraham silberschatz	wiley	2019	650	education
103	computer network	andrew s . tanenbaum	pearson	2020	720	education
104	data structure using c	reema thareja	oxford university press	2017	480	programming
105	artificial intelligence	stuart russell	pearson	2021	900	technology
106	python programming	gudiu van rossum	o reilly media	2022	550	programming
107	machine learning basics	tom m mitchell	mcgraw hill	2020	700	technology
204	computer networks	andrew s tanenbaum	pearson	2020	720	education
205	data structures and algorithms	thomas h cormen	mit press	2020	800	programming
206	artificial intelligence basics	stuart russell	pearson	2022	900	technology
301	software engineering	ian somerville	pearson	2021	850	education
302	web technologies	jeffrey ullman	pearson	2020	700	technology
303	introduction to algorithms	thomas h cormen	mit press	2019	950	programming
304	software engineering	ian somerville	pearson	2021	850	education
305	web technologies	jeffrey ullman	pearson	2020	700	technology
306	introduction to algorithms	thomas h cormen	mit press	2019	950	programming

16 rows in set (0.015 sec)

- Lab 4: Set a SAVEPOINT before making updates to the members table, perform some updates, and then roll back to the SAVEPOINT.

```
mysql> start transaction;
Query OK, 0 rows affected (0.009 sec)

mysql> update member set date_of_membership = "2022-12-20" where member_id=3;
Query OK, 1 row affected (0.109 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> update member set date_of_membership = "2025-4-12" where member_id=4;
Query OK, 1 row affected (0.015 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> savepoint s1;
Query OK, 0 rows affected (0.098 sec)

mysql> update member set date_of_membership = "2022-05-12" where member_id=5;
Query OK, 1 row affected (0.020 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> rollback to s1;
Query OK, 0 rows affected (0.093 sec)

mysql> select * from member;
+-----+-----+-----+-----+
| member_id | member_name | date_of_membership | email |
+-----+-----+-----+-----+
| 1 | Amit Patel | 2023-01-15 | amit.patel@gmail.com |
| 2 | Riya Shah | 2023-03-10 | riya.shah@gmail.com |
| 3 | Karan Mehta | 2022-12-20 | karan.mehta@gmail.com |
| 4 | Neha Joshi | 2025-04-12 | neha.joshi@gmail.com |
| 5 | Rahul Verma | 2023-09-18 | rahul.verma@gmail.com |
+-----+-----+-----+-----+
5 rows in set (0.007 sec)
```

SQL Joins LAB EXERCISES:

- Lab 3: Perform an INNER JOIN between books and authors tables to display the title of books and their respective authors' names.

```
mysql> select titles , author from books inner join author on author.author_id=books.book_id;
+-----+-----+
| titles | author |
+-----+-----+
| database management system | raghu ramkrishnan |
| operating system concept | abraham silberschatz |
| computer network | andrew s . tanenbaum |
| data structure using c | reema thareja |
| artificial intelligence | stuart russell |
+-----+-----+
5 rows in set (0.019 sec)
```

- Lab 4: Use a FULL OUTER JOIN to retrieve all records from the books and authors tables, including those with no matching entries in the other table.

```
mysql> select * from books left join author on author.author_id=books.book_id;
-> select * from books right join author on author.author_id=books.book_id;
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| book_id | titles | author | publish | year_of_publication | price | genre | author_id | first_name | last_name | country |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 181 | database management system | raghu ramkrishnan | mcgraw hill | 2018 | 556 | education | 281 | raghu | ramkrishnan | india |
| 182 | operating system concept | abraham silberschatz | Wiley | 2019 | 690 | education | 282 | abraham | silberschatz | usa |
| 183 | computer network | andrew s . tanenbaum | pearson | 2020 | 770 | education | 283 | andrew | tanenbaum | netherlands |
| 184 | data structure using c | reema thareja | oxford university press | 2017 | 380 | programming | 284 | reema | thareja | india |
| 185 | artificial intelligence | stuart russell | pearson | 2021 | 946 | technology | 285 | stuart | russell | usa |
| 186 | python programming | guidu van rossum | o reilly media | 2022 | 550 | programming | NULL | NULL | NULL | NULL |
| 187 | machine learning basics | tom mitchell | mcgraw hill | 2020 | 780 | technology | NULL | NULL | NULL | NULL |
| 188 | computer networks | andrew s tanenbaum | pearson | 2024 | 720 | education | NULL | NULL | NULL | NULL |
| 245 | data structures and algorithms | thomas h cornes | mit press | 2020 | 880 | programming | NULL | NULL | NULL | NULL |
| 246 | artificial intelligence basics | stuart russell | pearson | 2022 | 946 | technology | NULL | NULL | NULL | NULL |
| 382 | software engineering | ian somerville | pearson | 2021 | 880 | education | NULL | NULL | NULL | NULL |
| 383 | web technologies | jeffrey allman | pearson | 2020 | 780 | technology | NULL | NULL | NULL | NULL |
| 384 | introduction to algorithms | thomas h cornes | mit press | 2019 | 950 | programming | NULL | NULL | NULL | NULL |
| 385 | software engineering | ian somerville | pearson | 2021 | 850 | education | NULL | NULL | NULL | NULL |
| 386 | web technologies | jeffrey allman | pearson | 2020 | 780 | technology | NULL | NULL | NULL | NULL |
| 387 | introduction to algorithms | thomas h cornes | mit press | 2019 | 950 | programming | NULL | NULL | NULL | NULL |
16 rows in set (0.074 sec)
```

SQL Group By LAB EXERCISES:

- Lab 3: Group books by genre and display the total number of books in each genre.


```
mysql> select genre,count(titles) from books group by genre;
```

genre	count(titles)
education	6
programing	2
technology	5
programming	3

```
4 rows in set (0.109 sec)
```

- Lab 4: Group members by the year they joined and find the number of members who joined each year.

```
mysql> select date_of_membership , count(member_name) from member group by date_of_membership;
```

date_of_membership	count(member_name)
2023-01-15	1
2023-03-10	1
2022-12-20	1
2025-04-12	1
2023-09-18	1

```
5 rows in set (0.015 sec)
```

17. SQL View LAB EXERCISES:

Lab 3: Create a view to show only the title, author, and price of books from the books table.

```
mysql> create view book as select titles,author, price from books;
Query OK, 0 rows affected (0.074 sec)
```

- Lab 4: Create a view to display members who joined before 2020.

```
mysql> create view members1 as select * from member where date_of_membership < "2020-12-30";
Query OK, 0 rows affected (0.216 sec)
```

```
mysql> select * from members1;
Empty set (0.031 sec)
```